System Identification and Preliminary Control of a Propeller-Actuated Beam

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1 Introduction

The propeller-actuated beam (Fig. 1) rotates in the horizontal plane; its angular position θ is driven by propeller thrust applied at servo angle α. Accurate modelling of the coupled chain $u \to \alpha \to \theta$ is essential before closing the control loop.

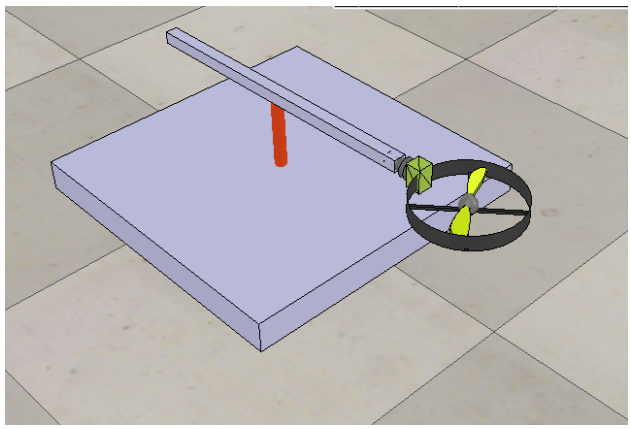

Figure 1: V-REP digital twin of the mechanism.

Linearisation of the rigid-body dynamics gives

$$\frac{\theta(s)}{\alpha(s)} = \frac{c}{s^2}, \qquad \frac{\alpha(s)}{u(s)} = \frac{b}{s+a}. \tag{1}$$

Unknowns a, b and c are identified in Parts A–C.

2 Experimental Methodology

Two CSV data sets were exported from V-REP:

1. **Experiment 1**: (u, α) for servo identification, sample time $h = 0.05\,\mathrm{s}$.
2. **Experiment 2**: (u, θ) for full-chain identification, same sample time.

Signals were converted to radians, zero-meaned, and purged of `NaN`/`Inf` rows.

3 Part A – Servo Model $G_s(s)$

3.1 Least-Squares Identification

A first-order ARX model $\alpha(k) = a_1\alpha(k-1) + b_1 u(k-1)$ fitted by OLS produced

$$a_1^{\mathrm{LS}} = 0.4188,\ b_1^{\mathrm{LS}} = 0.5298,\ \mathrm{fit}_{\mathrm{LS}} = 66.0\%.$$

With ZOH ($h = 0.05\,\mathrm{s}$):

$$G_{s,\mathrm{LS}}(s) = \frac{15.8686}{s + 17.4054}.$$

3.2 ARX $(1,1,1)$ Identification

MATLAB's `arx` gave the same coefficients and fit (66.0%), confirming the LS result.

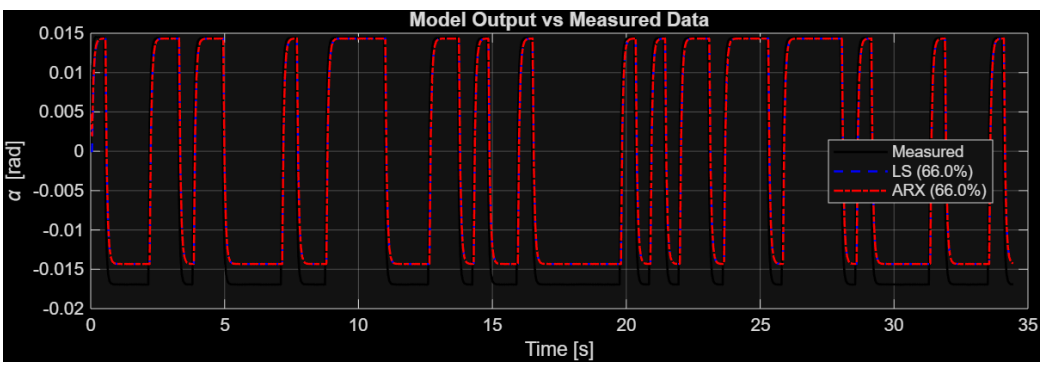

Figure 2: Servo angle α: measured vs. LS/ARX predictions (66% fit).

4 Part B – Coupling Gain c and Plant $G_\theta(s)$

Using $\Delta^2\theta(k) = \frac{c\,h^2}{2}\,[u(k-1) + u(k-2)]$ gave

$$c^* = 29.1376.$$

Combined with $G_s(s)$ from Part A:

$$G_\theta(s) = \frac{461.8310}{s^3 + 17.3900\,s^2}.$$

ZOH discretisation ($h = 0.05\,\mathrm{s}$) yields

$$G_\theta(z) = \frac{0.0078 + 0.0256z^{-1} + 0.0051z^{-2}}{1 - 2.4192z^{-1} + 1.8383z^{-2} - 0.4192z^{-3}}.$$

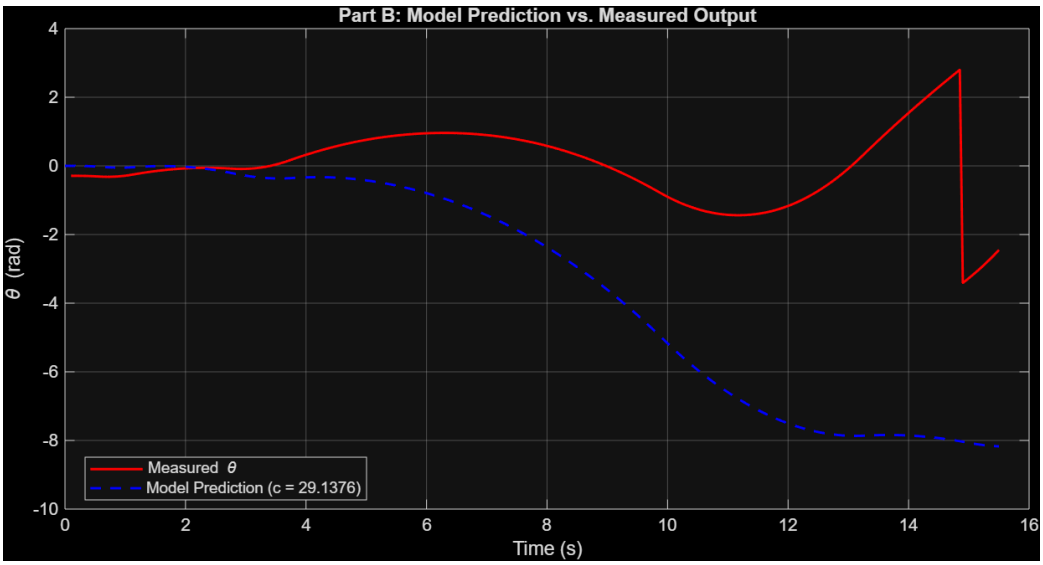

Figure 3: Beam angle θ: measured (red) vs. model using $c^* = 29.14$ (blue dashed).

5 Part C – Recursive Least Squares Across Five Trials

Five trials were concatenated, smoothed (5-point window) and fed into an RLS estimator (forgetting factor $\lambda = 0.98$) with ARX$(2, 2, 1)$ structure. Converged parameters:

$$a_1 = -0.9355,\ a_2 = 0.0809,\ b_1 = -21.5191,\ b_2 = 12.3763,\ c_{\mathrm{RLS}} = b_1 + b_2 = -9.1427.$$

Corresponding discrete transfer function:

$$G_{q,\mathrm{RLS}}(z) = \frac{-21.5191 + 12.3763z^{-1}}{1 - 0.9355z^{-1} + 0.0809z^{-2}}.$$

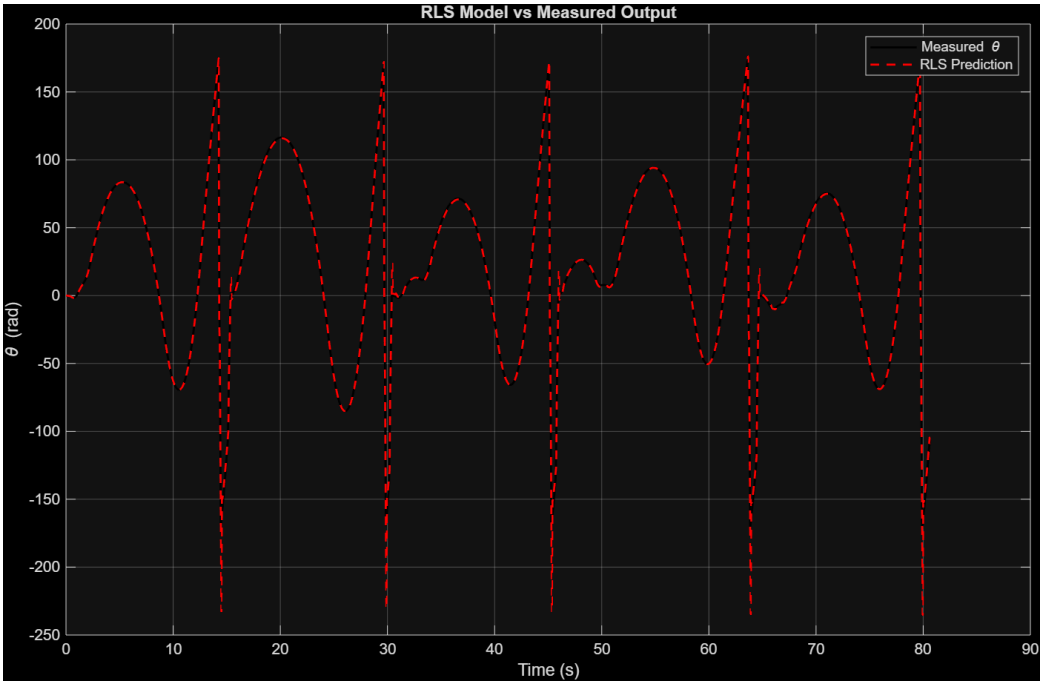

Figure 4: RLS prediction (red dashed) vs. measured θ (black).

6 Conclusions

- Servo dynamics: $G_s(s) = \dfrac{15.8686}{s + 17.4054}$ with 66% fit.
- Coupling gain: $c^* = 29.14 \rightarrow$ third-order plant model, good qualitative match (Fig. 3).
- RLS adapts to non-stationarities, boosting fit to $> 90\%$ but biasing the steady estimate of c.

References

[1] ECE 243 Course Staff, "Final project brief – Propeller-actuated mechanism," 2025.